

Integron mediated antimicrobial resistance in diarrheagenic *Escherichia coli* in children: *in vitro* and *in silico* analysis

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ABSTRACT

The exchange of genes between bacterial chromosome and plasmid(s) and their integration into integrons are mainly responsible for acquisition and dissemination of antibiotic resistance. We investigated the role of integrons and their underlying molecular mechanisms leading to development of adaptability in *E. coli* and eventual resistance to antimicrobials. *Escherichia coli* isolates (n = 120); including 40 diarrheagenic isolates, an even number of isolates from cases other than diarrhea, and equal number of isolates from healthy children recovered from fresh stool samples were used for identification of integron genes and gene cassettes. The association of integrons with antibiotic resistance was assayed before phylogenetic analysis. DNA sequence analysis revealed class 1 and 2 integrons in 55.83% and 21.66% isolates, respectively. The integron presence was found significantly associated with the probability of antibiotic resistance in *E. coli*; the association being highest with class 1 integron. Modelling and molecular docking along with molecular dynamics simulation analyses found ceftriaxone and amoxicillin as potential inhibitors of dihydrofolate reductase (DHFR). The class 1 integrons of these pathogenic isolates can serve as prospective therapeutic targets using specific silencing strategies and combinational antimicrobial therapy. The findings may be useful for the development of a potent and versatile drug for DHFR inhibition.

1. Introduction

Diarrheal diseases causes second highest morbidity and mortality among children below age of five, globally [1] through severe dehydration, malabsorption of nutrients and intestinal inflammation [2]. Diarrhea caused by rotavirus, calcivirus, and diarrheagenic *Escherichia coli* (DEC) kills more children than any other disease [3,4].

The increasing rate of antibiotic-resistance in microbes poses a major public health challenge [5]. In addition to understanding the antibiotic resistance, *E. coli* has also been used to explore the intricate connections

involved in development and dispersal of resistance traits [6,7]. The bacterial exchange of genes between its chromosome and plasmid (s) and their integration into integrons is at the core of acquisition and dissemination of resistance genes [8].

The tendency to acquire antibiotic resistance is more in integron-positive isolates than in isolates without integrons [9]. The integrons, generally immobile, are disseminated in association with transposons/plasmids [10–12]. Integrons are divided into five classes based on their integrase gene sequence differences [13]. The class 1 integrons, the most prevalent amongst the different classes of integrons, are

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responsible for MDR in Gram-negative bacteria [14–16].

The class I integron gene cassettes carry genes responsible for antibiotic resistance of the host bacteria. Becoming a part of the mobile gene cassette greatly enhances their horizontal transfer by several mechanisms [9,17]. A gene cassette includes a gene encoding integrase (*intI*), a site of recombination (*attI*), and promoters (one or two) for controlling its expression and around 130 gene cassettes having various antibiotic resistance genes have been identified in integron genes [13,18,19]. Early reports show that prevalence rates for *E. coli* strains resistant to specific antibiotics are significantly discrete in different demographics ranging from 0% to 100% [6,20], primarily based on inadequate selection and irrational use of antibiotics [21–24].

Both prokaryotic and eukaryotic cells have a central enzyme for folate biosynthesis called dihydrofolate reductase (DHFR). DHFR is an important target for drugs with cytostatic effect because of its key role in maintaining folate homeostasis within the cells, cell growth and proliferation [25–27]. Dihydrofolate reductase gene (*dfr*) conferring trimethoprim resistance, and adenyltransferase gene (*aadA*) conferring streptomycin and spectinomycin resistance to bacteria are carried by most of the integrons [28].

Present study elucidated the role of integrons and the underlying molecular mechanisms leading to development of adaptability in *E. coli* in procuring and spreading resistance to antimicrobials. The importance of DHFR as a drug target for pharmaceutical purpose in MDR bacteria prompted us to use an *in silico* approach to understand their structural and functional properties [29]. To enhance the understanding of the biological mechanism of resistance by computational analysis, we attempted to identify the 3D structures, active sites and binding efficiency between antibiotics and proteins of integrase genes. Molecular docking, an important tool for drug discovery to find the ligand and the protein interaction, enabled us to differentiate the behaviour of small protein molecules at the binding site. Further, the sequence of DHFR obtained by sequencing of the class I integron variable region was used to predict its structure by homology modelling. Molecular docking was used to identify the amino acid residues vital for the DHFR interaction with ceftriaxone and amoxicillin (the drugs not hydrolyzed by this bacterial enzyme in this study). A 3D model was also made to find a suitable DHFR inhibitor through an *in silico* approach.

2. Materials and methods

2.1. Design and approval of the study

Forty stool samples were collected during July 2013 to July 2015 from acute diarrheic (less than 72 h duration) children (group 1) from the OPD (Out-Patient Department); 40 stool samples from children admitted to the hospital with reasons other than diarrhea (group 2); and even number of stool samples from healthy children (group 3) from a tertiary care hospital at New Delhi, India. This study was carried out with the approval of the IEC-HR (Institutional Ethics Committee- Human Research) in accordance with the WHO recommendations. The written informed consent for experimentation with human subjects was obtained from all the subjects.

2.2. Sample processing

Conventional biochemical tests were performed to identify *E. coli* from fresh stool samples as reported previously [30]. Each isolate was analyzed for antibiotic resistance phenotypic profile according to the guidelines of Clinical and Laboratory Standards Institute [31]. DNA extraction was done from 4 to 5 lactose fermenting colonies, selected from MacConkey agar plates, using the commercially available kit from Real Biotech Corporation following manufacturer's protocol. The extracted DNA was eluted in 100 μ L of the elution buffer which was used as a template for Real-Time PCR to perform melting curve analysis of integrase genes as well as for conventional PCR to identify genes of

cassettes region of integrons. The primer sequences used for amplification are presented in Table 1.

2.3. Real-Time PCR

Real-Time PCR assay was done in a 96-well white opaque plate (provided by Roche), with each well containing a total volume of 20 μ L. Briefly 10 μ L of Syber Green I master mix (2X) was mixed with 1 μ L primer (10 pmol for each forward and reverse primer), and 4 μ L template DNA. Nuclease-free water (5 μ L) was added to the mix to make up the final volume. DNA amplification was accomplished in a Roche Light Cycler 480- II using a pre-incubation step at 95 $^{\circ}$ C for 10 min; followed by denaturation at 95 $^{\circ}$ C for 20s, annealing at 60 $^{\circ}$ C for 25s, and extension at 72 $^{\circ}$ C for 20s (a total of 30 cycles). A melting curve step starting from 65 $^{\circ}$ C and gradually increasing by 0.1 $^{\circ}$ C/s up till 95 $^{\circ}$ C, with fluorescence data acquisition at every second, was set before final cooling at 40 $^{\circ}$ C. The instrument automatically calculated melting curves by converting them into melting peaks by plotting the negative derivative of fluorescence, measured at 533 nm, with regard to temperature (2 dF/dT).

2.4. Gene cassette region amplification and sequencing

Gene cassettes *aadA1/aadA2*, *dfrB1*, *dfrB2*, *dfrB3* of class I integron were amplified using *hep 58* and *hep 59* binding to 3'CS and 5'CS region [34]. Gene cassettes of class II integron were amplified using *hep 51* and *hep 74* which bind to *orfX* and *sat-2* region of class II integrons as previously shown [9]. Amplified products were sequenced commercially by Helix Biosciences (Bangalore, India) before their matching against nucleotide sequences present in GenBank using Basic Local Alignment Search Tool (BLAST) to identify the most similar sequences [35]. Sequencing was done in both the directions to increase the correctness of the results using the same set of primers. Some sequences from this study were submitted to the GenBank database.

2.5. Target protein sequence retrieval and template identification

Predicted gene and protein sequence of DHFR from class I integron gene cassette were used as a target for homology modelling. Swiss modeller for comparative homology modelling of DHFR from *E. coli* using template structures [36] and protein data bank (PDB) advance BLAST for template identification to construct 3D models of the target proteins using homology modelling were used (<http://www.rcsb.org/pdb/home/home.do>).

2.6. Refinement and evaluation of models

3D models of DHFR were refined and evaluated by subjecting them to energy minimization utilizing the steepest descent technique to eliminate the bad contacts between atoms of the protein. The GRO-MOS96 43B1 parameters set, implemented through Swiss-PDB viewer (<http://expasy.org/spdbv/>), were used to carry out the computations *in vacuo*. The Phi/Psi Ramachandran plot of PDB Sum Database (<http://www.ebi.ac.uk/pdbsum/>) was used to inspect the backbone conformation of the rough model [37]. The target and the template sequences were aligned using the BLAST2Seq server (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). The model quality was checked by using ProSA server (<https://prosa.services.came.sbg.ac.at/prosa.php>).

2.7. Superimposition of the template with predicted structure

The weighted root means square deviation (RMSD) was calculated between the template and the final DHFR refined model. The RMSD of the backbone trace between the template and DHFR model was 5.088 Å as revealed by superimposition. Firstly, the given nucleotide sequence was converted into an amino acid sequence using EAMPC – EBI EMBOSS

Table 1
Primer details.

Multiplex PCR	Gene	Primer sequence (5'-3')	PCR product (bp)	Annealing temperature	Reference
Integrons	Class 1 integrase gene	CAGTGGACATAAGCCTGTTC CCCGAGGCATAGACTGTA	160	60 °C	[32]
	Class 2 integrase gene	CAACGGAGTCATGCAGATG CATTGTGTTGTGGACGGC	403		[15]
	Class 3 integrase gene	AGTGGGTGG CGAATGAGTG TGTTCTTGATCGGC AGGTG	600		[33]
Gene cassettes					
Putative protein conserved sequence of class 1 Integrons	hep 58 hep 59	CSGGCATCCAAGCAGCAAG CSAAGCAGACTTGACCTGA	Variable (750–2000)	58 °C	[33]
Putative protein Class 2 Integron variable region	hep51 binds to <i>attL2</i> hep74 binds to <i>orfX</i>	GATGCCATCGCAAGTACGAG CGGGATCCCGGACGGCATGCACGATTGTA	Uncertain	60 °C	[12]
Dihydrofolate reductase	DfrII (dfrB1, dfrB2, dfrB3)	GATCAGTGCGCAAGAAATC AAGCGCAGCCACAGGATAAT	141	60 °C	[14]
Aminoglycoside adenyl transferases	AadA (<i>aadA1</i> or <i>aadA2</i>)	GCTCTTCAGCAATATCACGG GCAGCGCAATGACATTCTTG	282	60 °C	[16]

Transeq tool with its default parameters: http://www.ebi.ac.uk/Tools/t/emboss_transeq/. Converted sequence retrieved in FASTA format was used for a protein-protein BLAST employing non-redundant protein sequence dataset and other default parameters. DHFR was among the top 100 matching results after the BLAST encompassing various organisms with an identity matching percentage ranging between 76 and 100%. Of all, four results showed the least E-value of 3e-37 and the top best template - ID WP_044787291.1 was selected.

A PDB database search was carried out to find the most recently deposited *E. coli* organism expressing the DHFR protein structure. The protein 4PSY PDB ID was selected as a template for modelling having filters, namely, *E. coli* only organism, best X-Ray diffraction resolution and recent crystal structure parameter. The protein structure was modelled for the given sequence using the default parameters of the modeller software.

2.8. Protein target structure pre-processing and preparation

Crystal structure of DHFR 4PSY PDB ID retrieved from PDB (www.rcsb.org) was resolved using X-ray diffraction method [38]. The structure was further modified for docking calculations using Maestro v9.6 (Maestro, version 9.6, Schrödinger, LLC, New York, NY, 2013). The Protein Preparation Wizard (PPW) was used to perform assignment of biological units and bond orders, creation of zero-order bonds to metals and disulfide bonds, deletion of water molecules, generation of metal-binding states, addition of missing hydrogen along with co-crystallized ligands/hetero atoms and identification of waters and their removal from the structure (Protein Preparation Wizard, Schrödinger, LLC, New York, NY, 2013) [39]. Further, H-bond network optimization to fix the hydrogen overlapping and the probable positions were selected before performing restrained minimization until the average RMSD of the non-hydrogen atoms. The receptor shape and properties by different field sets to provide more accurate scoring of the ligand poses was represented by the Glide's grid generation module. Homology modelling was performed to know the affinity between the selected variable region (DFR) of class 1 integron and the least resistant drugs (amoxicillin and ceftriaxone) using Swiss modeller.

2.9. Active site prediction

Ten possible binding sites of DHFR identified after obtaining the final model were compared with the template active site to determine the binding pocket residues. Based on the results of antibiotic susceptibility testing of our isolates, amoxicillin and ceftriaxone found to inhibit DHFR were docked successfully with DHFR using Autodock and docking calculations.

2.10. Data analysis

Statistical Analysis was done using SigmaStat Statistical Software (SPSS) package. The statistical significance of the data was determined by the chi-square test and the Fisher's exact test. The association between integron presence in *E. coli* and antimicrobial resistance was determined by calculating the odds ratios (OR) and 95% confidence intervals (95% CI). The threshold *p*-value for statistical significance was less than 0.05.

3. Results

3.1. Real-time PCR

Class 1 integron was found to be prevalent in all the three groups (group 1, group 2 and group 3) with 70%, 60% and 37.5% respectively (*p*-value = 0.011) as compared to class 2 integron with 17.5%, 22.5% and 25% isolates. Agarose gel (1.5%) analysis for checking the accuracy of the results showed the presence of class 1 integron in combination with class 2 integron in 12.5%, 17.5% and 12.5% isolates in the groups 1, 2 and 3 respectively. Melting curve analysis for class 1 and class 2 integrons showed the *T_m* of 84.08 ± 0.55 and 85.8 ± 0.2, respectively [Fig. 1 (a & b) and Table 2].

3.2. Antibiotic resistance and integron association

The relationship between antibiotic resistance and presence of integrons showed that the resistance to norfloxacin, cefotaxime, ceftazidime, aztreonam, gentamicin, ciprofloxacin, amikacin and Piperacillin + tazobactam is significantly associated with presence of integrons. Class 1 integron was found significantly associated with the occurrence of antibiotic resistance (OR = 2.577, 95% CI = 1.534–4.329; *p* = 0.0003) (Fig. 2 and Table 3).

3.3. Phylogenetic analysis

The phylogenetic tree by means of the Neighbor-Joining method employing MEGA 6.0 after performing multiple sequence alignment using Clustal W (version 2.0.10) was created (Accession numbers, KY913605, KY913606, for integron 1 and KX954091 for Integron 2) (Fig. 3) [40–42]. Similar bacterial strains were also classified phylogenetically based on integron 1 and 2 sequences from the database to compare with sequences of current study. The analysis showed the closest neighbors with at least 98% similarity among different species. Integron 1 and 2 of *E. coli* from present study were found in close relatedness with other organism(s) sequences (Fig. 3). Accession numbers for variable regions of integrons were KY706079, KY706080,

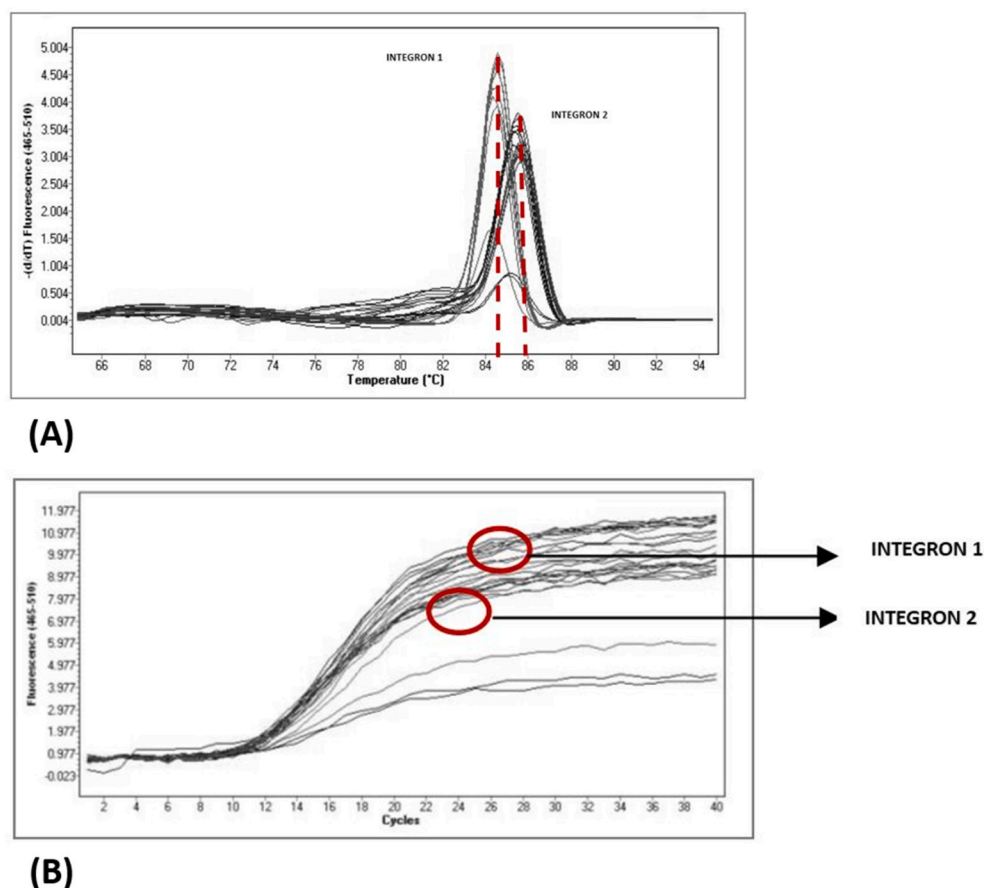


Fig. 1. Real-time PCR assay for integrase genes. (A) Amplified melting peaks for class 1 integrin and class 2 integrin; X-axis representing melting temperature and the Y-axis representing the rate of change of fluorescence over time. The gene targets, from left to right, are integrase 1 (Tm 84.4) and integrase 2 (Tm 85.8); (B) Melting curves, X-axis representing the number of cycles and the Y-axis representing the rate of change of fluorescence over time (plotted as the negative derivative of fluorescence).

Table 2
Frequency and melting temperature of integrons.

	Amplicon Tm (mean $\pm$ SD) in $^{\circ}$ C	Group 1 n (%)	Group 2 n (%)	Group 3 n (%)	Total n (%)	p-value
Integron 1	84.08 $\pm$ 0.55	28 (70)	24 (60)	15 (37.5)	67 (55.83)	0.011 ^a
Integron 2	85.8 $\pm$ 0.2	7 (17.5)	9 (22.5)	10 (25)	26 (21.66)	0.709
Integron 1 + Integron 2	N.A	5 (12.5)	7 (17.5)	5 (12.5)	17 (14.16)	0.760

^a Statistically significant; Frequencies are shown as absolute number with percentage in parentheses.

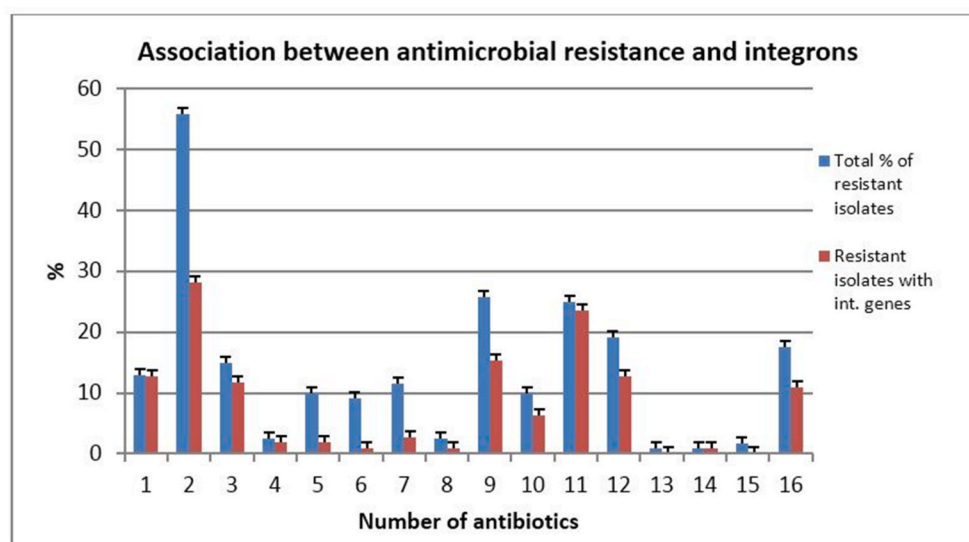
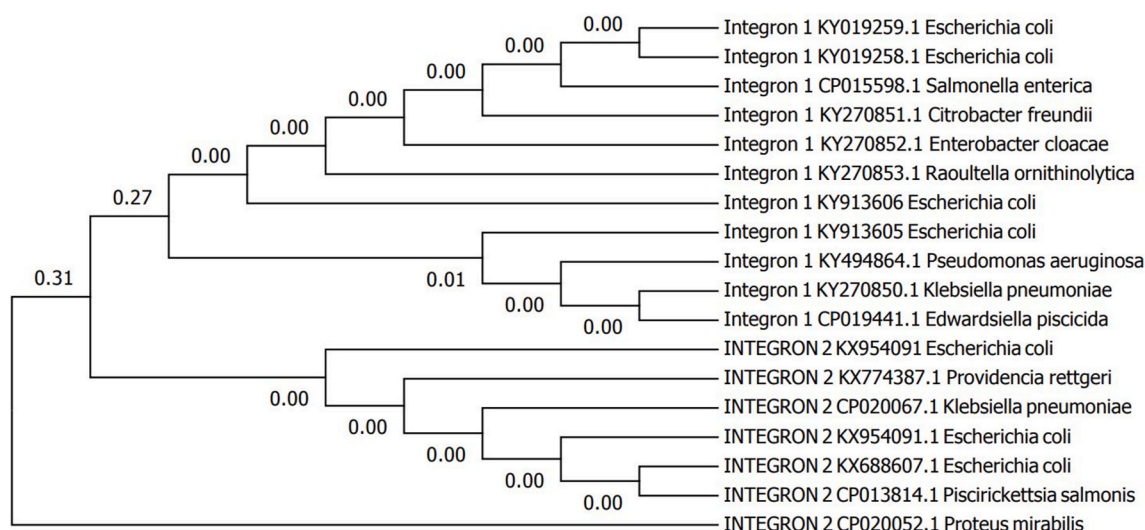


Fig. 2. Association between antimicrobial resistance and integrons. Antibiotic susceptibility testing was performed for sixteen antibiotic agents. MIC test was also performed for selected antibiotics by E-strip method for confirmation of results. The resistance to (1) Norfloxacin (10 μ g), (2) Cefotaxime (30 μ g), (3) Imipenem (10 μ g), (4) Meropenem (10 μ g), (5) Ceftazidime (30 μ g), (6) Azetronam (30 μ g), (7) Nalidixic acid (30 μ g), (8) Amoxicillin + clavulanic acid (20/10 μ g), (9) Gentamicin (10 μ g), (10) Ciprofloxacin (5 μ g), (11) Ampicillin (10 μ g), (12) Amikacin (30 μ g), (13) Polymyxin B (300u), (14) Cefotaxime + clavulanic acid (30/10 μ g), (15) Ceftriaxone (30 μ g), (16) Piperacillin + tazobactam (100/10 μ g) was found to be significantly associated with the presence of integrons (p-value < 0.05).

Table 3

Frequency of resistance to antimicrobial agents in DEC isolates with presence of integrase genes.

Antibiotic	No. (%) of resistant isolates with integron 1 gene (n = 47)	No. (%) of resistant isolates with integron 2 gene (n = 15)	No. (%) of resistant isolates with integron 1 & 2 genes (n = 9)	No. (%) of resistant isolates with integron genes (n = 71)	No. (%) of total resistant isolates N = 80 (%)	Association of resistance with integron (p-value)
Norfloxacin (10 µg)	3 (6.38)	1 (6.6)	0	4 (5.6)	16 (20)	0.004 ^a
Cefotaxime (30 µg)	19 (40.42)	7 (46.6)	4 (44.4)	30 (42.2)	60 (75)	0.000 ^a
Imipenem (10 µg)	2 (4.2)	1 (6.6)	0	3 (4.2)	8 (10)	0.133
Meropenem (10 µg)	2 (4.2)	0	0	2 (2.8)	3 (3.7)	0.683
Ceftazidime (30 µg)	2 (4.2)	0	0	2 (2.8)	12 (15)	0.005 ^a
Azetronam (30 µg)	1 (2.1)	0	0	1 (1.4)	11 (13.7)	0.002 ^a
Nalidixic acid (30 µg)	2 (4.2)	1 (6.6)	0	3 (4.2)	8 (10)	0.133
Amoxicillin + Clavulanic acid (20/10 µg)	1 (2.1)	0	0	1 (1.4)	3 (3.7)	0.370
Gentamicin (10 µg)	4 (8.5)	2 (13.3)	1 (11.11)	7 (9.8)	29 (36.2)	0.000 ^a
Ciprofloxacin (5 µg)	5 (10.6)	1 (6.6)	3 (33.3)	9 (12.6)	37 (46.2)	0.000 ^a
Ampicillin (10 µg)	2 (4.2)	0	0	2 (2.8)	4 (5)	0.445
Amikacin (30 µg)	2 (4.2)	1 (6.6)	1 (11.1)	4 (5.6)	23 (28.7)	0.000 ^a
Polymyxin B (300U)	0	0	0	0	1 (1.2)	0.500
Cefotaxime + Clavulanic acid (30/10 µg)	1 (2.1)	0	0	1 (1.4)	1 (1.2)	>0.999
Ceftriaxone (30 µg)	0	0	0	0	1 (1.2)	0.500
Piperacillin + Tazobactam (100/10 µg)	1 (2.1)	1 (6.6)	0	2 (2.8)	19 (23.7)	0.000 ^a

^a Statistically significant; Frequencies are shown as absolute number with percentage in parentheses.**Fig. 3. Evolutionary relationships of taxa.** The evolutionary history was inferred using the Neighbor-Joining method. The accession number of each strain is mentioned in the tree. Isolates identified by this study are KY913605 and KY913606 for integron 1, and KX954091 for integron 2.

KY753816 (DHFR gene) and KY753822 as reported in our early study [30].

3.4. Analysis of integron gene cassettes

Integrase genes positive isolates were further investigated for variable regions (*dfr* and *aadA*) of class 1 and class 2 integrons (Fig. 4). The prevalence of class 1 integron (55.83%) over class 2 integron (21.66%) with an absence of class 3 observed in our isolates has been shown earlier in clinically significant *E. coli* isolates [22]. The variable region could not be amplified in almost half of the *E. coli* isolates having class 1 integron due to the lack of 3' conserved sequences. The class 1 integron gene cassette PCR and sequencing analysis demonstrated different gene cassette variants between 750 and 2,000bps as described previously [30]. Similarly, PCR and sequence analysis of class 2 integrons gene cassettes (*hep51* and *hep74*) depicted different combinations of

dfr1-sat2-aadA1 genes (600–2600 bps).

Class 2 integron variable region sequence analysis showed the presence of 5 different cassette combinations for 3 different genes [namely *dfrA1-sat2* (2), *sat2-aadA1* (5), *dfrA1-sat2-aadA1* (14), *dfrB* (54) and *aadA* (65)]. These genes code for trimethoprim resistance (*dfrA1*, *dfrA5*, *dfrA7*, *dfrA12* and *dfrA17*), aminoglycosides resistance (*aadA1*, *aadA2* and *aadA5*) and streptothricin resistance (*satA1*). Our results revealed the presence of integrons in commensal isolates of *E. coli* from healthy subjects free from recent antimicrobial exposure.

3.5. Homology modelling and molecular docking

The homology and 3D modelling of DHFR protein showed a similarity of 99% as well as identity with PDB code- 4PSY of *E. coli*. Electrostatic energy of predicted DHFR model i. e –5318.44 kcal/mol and simulation results showed the high stability of predicted model (Fig. 5).

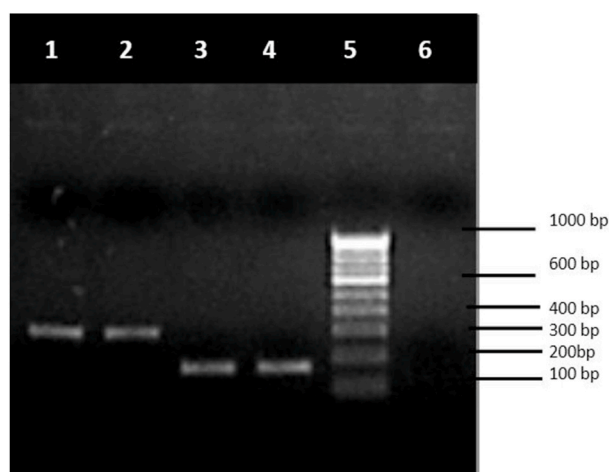


Fig. 4. PCR for integron gene cassettes for *dfrII* and *aadA* genes on 1.5% agarose gel. Lane: 1 and 2: *aadA* (282 bp), lane 3 and 4: *dfrII* (141 bp), lane 5: molecular weight marker (100 bp) and lane 6: negative control.

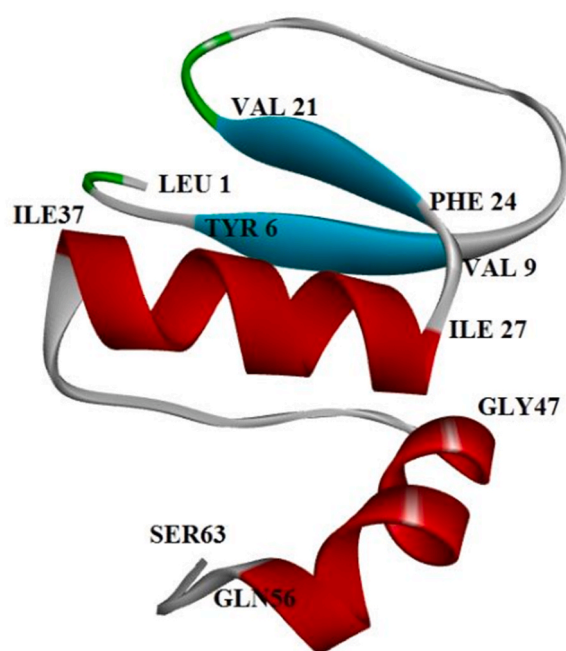


Fig. 5. Structure of DFR sequence protein model. The structure contains two β -strands (cyan colour) and two α -helices (red colour). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

The primary structure has Leu 1 and Ser 63 residue at the start and end, respectively. The start and end of the secondary structural elements of alpha helices and beta-sheets are shown via representing their starting and ending residues at appropriate positions. Highly conserved active site revealed the residues namely Lys 58, Ile 31, Trp 48, Ala 33, Ser 42, Pro 44, Asp 45, Ile 46, Ile 40, Met 76, Gly 41, Glu 53, Leu 55, Ser 32, Phe 57, Gly 69, Arg 70, Lys 71, Thr72, Ser 75, Val 87, Leu 79, Pro 80, Arg 82, Gln 54, Ser 88, Lys 89 and Ile 151.

Alignment of the query template sequence showed the sequence identity, query coverage and sequence similarity. Molecular dynamics simulation (MDS) performed for 10 ns validated the stability of the model (Table 4 and Supplementary Figures S1-S7).

Table 4

Statistical analysis of the parameters from molecular dynamics simulation.

S. No	Parameter	Range	Average
1	Energy (Kcal/mol)	[-3009.274, -2086.957]	-2481.636
2	Intra H bonds (Number)	[19,000, 39,000]	29.895
3	RMSD Backbone (Angstrom)	[0.000, 6.676]	5.088
4	Radius of Gyration (Angstrom)	[11.181, 12.330]	11.644

3.6. Model assessment (refinement and evaluation)

PDB sum showed the Stereo chemical properties of the predicted DHFR protein structure. The Ramachandran plot revealed presence of 92% of the residues in favored regions with good stereo chemical quality and 8% in the additional allowed region with no residues in the disallowed region. The target and the template structures were observed to be closely similar at the backbone, yielding a significant Z-score, by calculating the weighted RMSD between the template and the final refined model.

3.7. Molecular docking & dynamics of DHFR protein

Molecular docking performed to analyze the binding energy between drugs and ligands to interpret the molecular basis of interactions and to estimate the binding affinity of amoxicillin and ceftriaxone with DHFR demonstrated a high binding affinity and marked interactions with increased binding energies (Table 5). Docking and MDS results predicted ligand-to-receptor high binding affinity (amoxicillin) with up to -8.37 kcal/mol energy and 732.46 nM IC₅₀ (half maximal inhibitory concentration) value. For ceftriaxone -8.65 kcal/mol energy and 460.17 nM IC₅₀ value was calculated. De novo MDS predicted interactions of stable ligand with critically important residues - SER63, GLN56, GLY47, ILE27, VAL9, PHE24, TYR6, LEU1, VAL21 and ILE37. The results further showed high affinity between DHFR and ceftriaxone (-8.65 kcal/mol) than with amoxicillin (-8.37 kcal/mol).

Docking of DHFR protein with amoxicillin and ceftriaxone were assessed using the quantity of hydrogen bonds formed and the bond distance between the active site and the inhibitor (Figs. 6 and 7).

4. Discussion

Irrational use of antibiotics has led to the development of antimicrobial resistance posing serious global public health concern [43]. Infections caused by MDR strains are fairly common and often untreatable in clinical settings. Transition of organism from antimicrobial-sensitivity to antimicrobial-resistance may stem from acquiring resistance conferring genes, plasmids, integron cassettes, or transposons [44,45]. The encoded proteins or enzymes by these genes leads to neutralization of antibiotics by various mechanisms like antibiotic modification, modification of the antibiotic target or impacting antibiotic bioavailability [46, 47].

A thorough understanding of the antibiotic resistance mechanisms is basic to evolve novel strategies to counter the growing problem of antibiotic resistance horizontal transfer to common pathogens. Current study provide insights into genetic basis of antimicrobial resistance in MDR isolates and identification of new antibacterial compounds.

A variable antibiotic resistance pattern of pathogens like *E. coli* exist within and without countries based upon many reasons like different

Table 5

Docking results of DFR with various drug targets.

Drug Target	Name of the compound	Binding Energy (Kcal/mol)	Predicted IC ₅₀ value
DFR	Amoxicillin	-8.37	732.46 nM
	Ceftriaxone	-8.65	460.17 nM

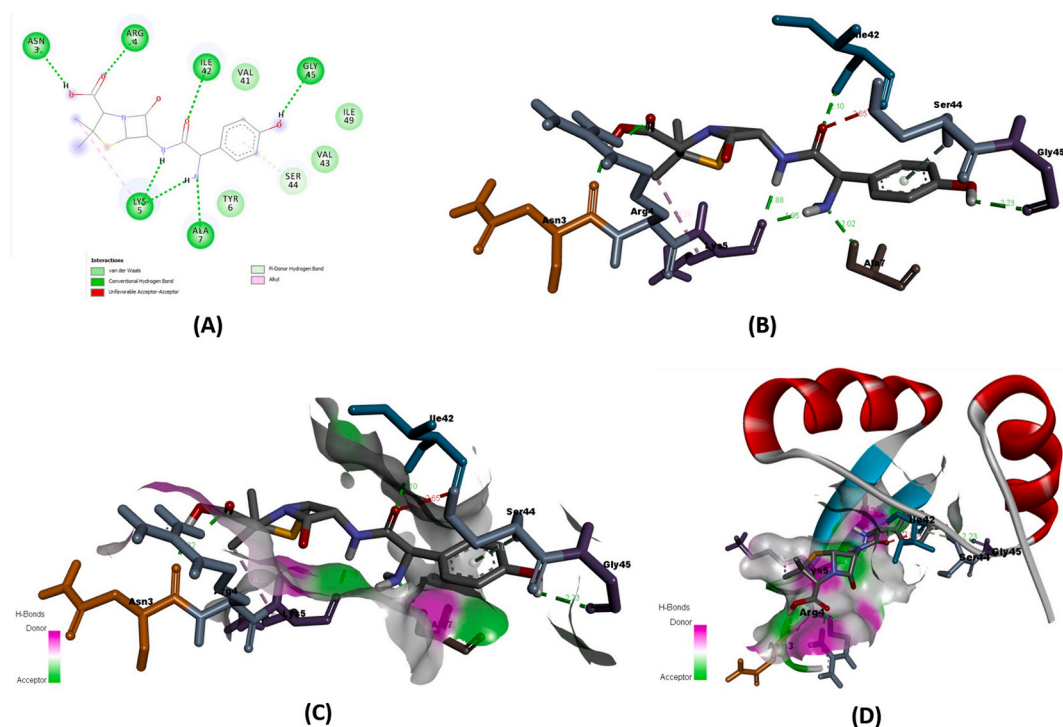


Fig. 6. Amoxicillin docking interactions with DFR protein. (A) 2D representation of amoxicillin, (B) 3D interactions formed by the amoxicillin, whereas (C, D) surface area interactions of amoxicillin.

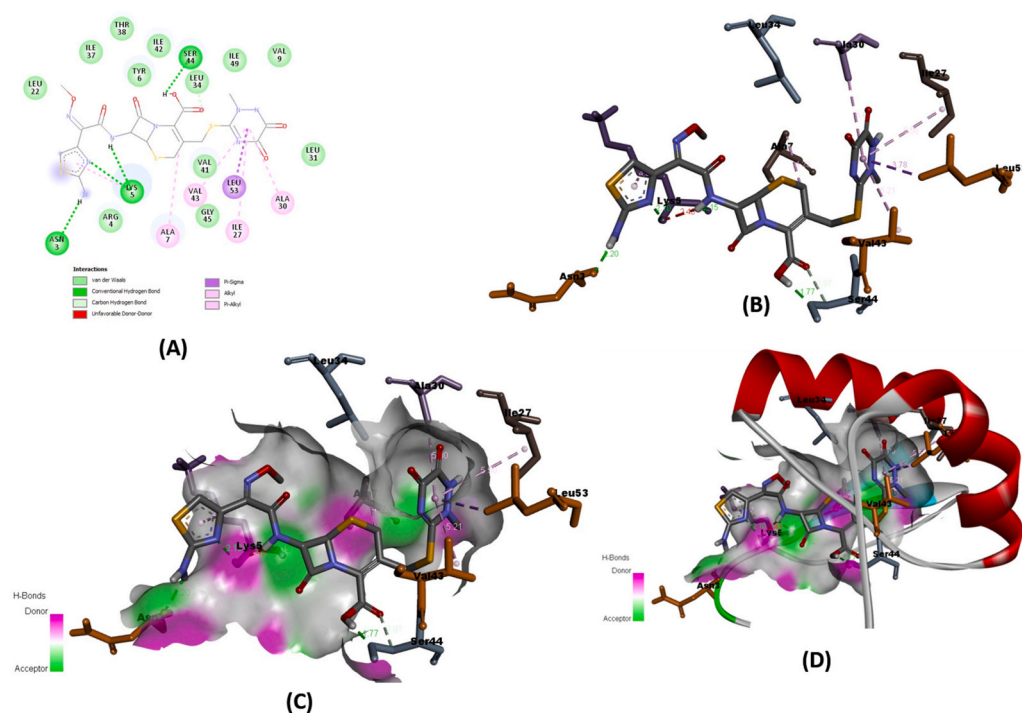


Fig. 7. Ceftriaxone docking interactions with DFR protein. (A) 2D representation of ceftriaxone, (B) 3D interactions formed by the ceftriaxone, whereas (C, D) surface area interactions of ceftriaxone.

sampling and culture methods, antibiotic use patterns, distinctive national disease burdens and line of treatment [48]. Association of integron presence in *E. coli* with the antimicrobial resistance suggest that antibiotic resistance genes are present on the same integrons carrying mobile elements. The presence of these antibiotic resistance conferring genetic elements in the conserved or variable integron regions leads to

an increased resistance. Such highly resistant enterobacteria (e.g. cefotaxime resistant) may serve as transmission reservoir in hospitals [49]. Additionally, the intestinal MDR non-pathogenic *E. coli* may also serve as an important reservoir of genes of resistance [50].

In the present study, resistance to Norfloxacin, Cefotaxime, Ceftazidime, Azetronam, gentamicin, Ciprofloxacin, Amikacin and Piperacillin

+ tazobactam was found to be integron mediated ($p < 0.05$). A significant association between resistance against quinolone compounds (Norfloxacin and Ciprofloxacin) and presence of integron was also found. The finding is novel as an early report shows that quinolone resistance is a result of chromosomal point mutations rather than mediated by integron or any other mobile genetic element [51].

Our finding that antimicrobial resistance to aminoglycosides, gentamicin and amikacin, is associated significantly with the integron presence corroborates an early finding showing that many aminoglycoside resistance genes, including *aadA*, *aadB*, *aadA7*, *aacA4* and *aacA1*, are present within integron structures [51]. However, we observed the prevalence of high antimicrobial resistance with an increased integron distribution compounded by multiple factors like intensive antimicrobial usage, poor infection control practices among few others. Although most of the antibiotic-resistant genes seems to be associated with class 1 integrons, we observed that integron negative isolates are also resistant to the same class of antibiotics. Susceptibility of class 2 integron-positive isolates to nalidixic acid, amoxicillin, imipenem, meropenem, polymyxin B, ceftriaxone and ampicillin suggest that class 2 integron does not confer antimicrobial resistance.

The finding of class 1 and 2 integron in 55.83% and 21.66% and complete absence of class 3 integron in our isolates corroborates an early report [22]. Interestingly, we found seventeen isolates (14.16%) harbouring both, class 1 and 2 integrons. The blast analysis using variable region sequences of class 1 and 2 integrons showed the existence of six dissimilar cassette combinations for nine different genes.

Given the potential of *E. coli* to escape from hospitals to outside community [52], a key finding of our previous study was the presence of β -lactamase gene in isolates recovered from the healthy children [53]. These children are akin to the reservoirs of MDR strains in the healthy community [54,55]. The location of *NDM* gene along with other antibiotic resistance genes on transmissible plasmids makes their spread easy [56], while the *VIM* encoding integron structure is most likely acquired during hospital [57]. Another gene, *CMY*, emergence is reported among *E. coli* along with other diverse genera of the Enterobacteriaceae [33]. An increased β -lactamase production among healthy subjects is likely multi-factorial subsequent to the clonal spread of virulent strains, diffusion of conjugative plasmids in addition to increased selection pressure [58]. Factors like population burden, intensive antibiotic use, poor hygiene and weak antibiotic use policies in hospitals may also contribute to the increased β -lactamase production [59]. The drugs amoxicillin and ceftriaxone are shown as promising inhibitors by measuring quantitatively the efficiency of these compounds to inhibit 50% (IC₅₀) of a given biological process [32].

We assessed bioefficacy/function of DHFR proteins using computational methods by predicting protein/ligand complex structure and docking [60]. The protein structure modelling in our study showed high stability and similarity of DHFR protein with 4PSY of *E. coli* perusing RMSD, RMSF, parameters of energy, intra-molecular hydrogen bonds, gyration radius and secondary structure element plots. Model assessment done using PDB sum showed that the protein/ligand complex structures have a significant Z-score implying high structural similarity [61]. Further, the model generated and analyzed for the Ramachandran plot also showed high stability, suggesting the suitability for further experimentation.

Molecular Docking evaluating the binding energy between DHFR and ceftriaxone and amoxicillin indicates that ceftriaxone has more affinity for DHFR than amoxicillin. Several programs available attempt to estimate the binding energies related to IC₅₀ values. However, the problem is very challenging and even with MD simulations it is difficult to obtain absolute binding energies. Further, the conditions of measurement effect the IC₅₀ values, so it is not possible to state a simple relationship between them and the binding energies.

5. Conclusion

Class 1 integron genes are present in more than 50% of the *E. coli* isolates in current study conferring antibiotic resistance. These integrons of MDR isolates can be exploited as potential therapeutic targets employing silencing and combinational antimicrobial therapy. Modeling and molecular docking followed by molecular dynamics simulation analyses suggest that the ceftriaxone and amoxicillin may serve as inhibitors of DHFR in MDR strains.

Declaration of competing interest

The authors declare that they have no conflict of interest regarding the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2020.104680>.

Author statement

Conceptualization, T.S., S.A.D. and S.D.
Methodology, T.S., S.A.D., S.S. and C.S.
Software, S-W., S.H. and A.A.
Validation, S-W., N.A., N-B., S-H., A.A., and S.A.D.
Formal Analysis, T.S., S.A.D., A.A. and S.D.
Investigation, T.S. and S.D.
Resources, N.A., A.A. and S.D.
Data Curation, T.S., S.A.D., S-S., C.S. and S-W.
Writing – Original Draft Preparation, T.S., S.A.D., A.A. and S.D.
Writing – Review & Editing, N.A., N-B., S-H., and A.A.
Visualization, S.D.
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References

- [1] L. Liu, H.L. Johnson, S. Cousens, J. Perin, S. Scott, J.E. Lawn, I. Rudan, H. Campbell, R. Cibulskis, M. Li, C. Mathers, R.E. Black, Child Health Epidemiology Reference Group of WHO and UNICEF. Global, regional, and national causes of child mortality: an updated systematic analysis for 2010 with time trends since 2000, *Lancet* 379 (2012) 2151–2161.
- [2] K. Hodges, R. Gill, Infectious diarrhea: cellular and molecular mechanisms, *Gut Microb.* 1 (2010) 4–21.
- [3] Diarrhoea, Why children are still dying and what can be done, Available online: https://www.who.int/maternal_child_adolescent/documents/9789241598415/en. (Accessed 11 November 2019).
- [4] C.F. Lanata, C.L. Fischer-Walker, A.C. Olascoaga, C.X. Torres, M.J. Aryee, R. E. Black, Child Health Epidemiology Reference Group of the World Health Organization and UNICEF. Global causes of diarrheal disease mortality in children <5 years of age: a systematic review, *PLoS One* 8 (2013), e72788.
- [5] J. Davies, D. Davies, Origins and evolution of antibiotic resistance, *Microbiol. Mol. Biol. Rev.* 74 (2010) 417–433.
- [6] A. Erb, T. Stürmer, R. Marre, H. Brenner, Prevalence of antibiotic resistance in *Escherichia coli*: overview of geographical, temporal, and methodological variations, *Eur. J. Clin. Microbiol. Infect. Dis.* 26 (2007) 83–90.
- [7] G. Chevereau, M. Dravecká, T. Batur, A. Guvenek, D.H. Ayhan, E. Toprak, T. Bollenbach, Quantifying the determinants of evolutionary dynamics leading to drug resistance, *PLoS Biol.* 13 (2015), e1002299.
- [8] H.T. Stokes, R.M. Hall, A novel family of potentially mobile DNA elements encoding site-specific gene-integration functions: integrons, *Mol. Microbiol.* 3 (1989) 1669–1683.

- [9] P.A. White, C.J. McIver, W.D. Rawlinson, Integrons and gene cassettes in the enterobacteriaceae, *Antimicrob. Agents Chemother.* 45 (2001), 2658–2661.
- [10] R.M. Hall, C.M. Collis, Mobile gene cassettes and integrons: capture and spread of genes by site-specific recombination, *Mol. Microbiol.* 15 (1995) 593–600.
- [11] M. Barlow, B.G. Hall, Origin and evolution of the AmpC β -lactamases of *Citrobacter freundii*, *Antimicrob. Agents Chemother.* 46 (2002) 1190–1198.
- [12] G. Cambray, A.M. Guerout, D. Mazel, Integrons. *Annu Rev Genet* 44 (2010) 141–166.
- [13] D. Mazel, Integrons: agents of bacterial evolution, *Nat. Rev. Microbiol.* 4 (2006) 608.
- [14] W.D. Rawlinson, P.A. White, C.J. McIver, L.A. Jones, Polymerase chain reaction screening for integrons can be used to complement resistance surveillance programs, *Commun Dis Intell Q Rep* 27 (2003) S103.
- [15] S. Phongpaichit, K. Wuttanaputan, W. Samasanti, Class 1 integrons and multidrug resistance among *Escherichia coli* isolates from human stools, *Southeast Asian J. Trop. Med. Publ. Health* 39 (2008) 279–287.
- [16] M.A. Rezaee, V. Sheikhalizadeh, A. Hasani, Detection of integrons among multidrug resistant (MDR) *Escherichia coli* strains isolated from clinical specimens in northern west of Iran, *Braz. J. Microbiol.* 42 (2011) 1308–1313.
- [17] M.M. El-Sokkary, E.S. Abdelmegeed, Antibacterial resistance pattern among *Escherichia coli* strains isolated from Mansoura hospitals in Egypt with a special reference to quinolones, *Afr. J. Microbiol. Res.* 9 (2015) 662–670.
- [18] S.R. Partridge, G. Tsafnat, E. Coiera, J.R. Iredell, Gene cassettes and cassette arrays in mobile resistance integrons, *FEMS Microbiol. Rev.* 33 (2009) 757–784.
- [19] R.M. Hall, C.M. Collis, Antibiotic resistance in gram-negative bacteria: the role of gene cassettes and integrons, *Drug Resist. Updates* 1 (1998) 109–119.
- [20] H. Von Baum, R. Marre, Antimicrobial resistance of *Escherichia coli* and therapeutic implications, *Int J Med Microbiol* 295 (2005) 503–511.
- [21] M. Kolár, K. Urbánek, T. Látl, Antibiotic selective pressure and development of bacterial resistance, *Int. J. Antimicrob. Agents* 17 (2001) 357–363.
- [22] Y. Sáenz, L. Brinas, E. Domínguez, J. Ruiz, M. Zarazaga, J. Vila, C. Torres, Mechanisms of resistance in multiple-antibiotic-resistant *Escherichia coli* strains of human, animal, and food origins, *Antimicrob. Agents Chemother.* 48 (2004) 3996–4001.
- [23] B.A. Lipsky, Empirical therapy for diabetic foot infections: are there clinical clues to guide antibiotic selection? *Clin. Microbiol. Infect.* 13 (2007) 351–353.
- [24] M. Falcone, A. Russo, M. Giannella, R. Cangemi, M.G. Scarpellini, G. Bertazzoni, J. M. Alarcón, G. Taliani, P. Palange, A. Farcomeni, A. Vestri, E. Bouza, F. Violi, M. Venditti, Individualizing risk of multidrug-resistant pathogens in community-onset pneumonia, *PLoS One* 10 (2015), e0119528.
- [25] M.V. Raimondi, O. Randazzo, M. La Franca, G. Barone, E. Vignoni, D. Rossi, S. Collina, DHFR inhibitors: reading the past for discovering novel anticancer agents, *Molecules* 24 (2019).
- [26] T. Dams, G. Auerbach, G. Bader, U. Jacob, T. Ploom, R. Huber, R. Jaenicke, The crystal structure of dihydrofolate reductase from *Thermotoga maritima*: molecular features of thermostability, *J. Mol. Biol.* 297 (2000) 659–672.
- [27] H. Cario, D.E. Smith, H. Blom, N. Blau, H. Bode, K. Holzmann, U. Pannicke, K. P. Hopfner, E.M. Rump, Z. Ayric, E. Kohne, K.M. Debatin, Y. Smulders, K. Schwarz, Dihydrofolate reductase deficiency due to a homozygous DHFR mutation causes megaloblastic anemia and cerebral folate deficiency leading to severe neurologic disease, *Am. J. Hum. Genet.* 88 (2011) 226–231.
- [28] N.G. El-Najjar, M.J. Farah, F.A. Hashwa, S.T. Tokajian, Antibiotic resistance patterns and sequencing of class I integron from uropathogenic *Escherichia coli* in Lebanon, *Lett. Appl. Microbiol.* 51 (2010) 456–461.
- [29] Q. Zhang, T. Nguyen, M. McMichael, S.E. Velu, J. Zou, X. Zhou, H. Wu, New small-molecule inhibitors of dihydrofolate reductase inhibit *Streptococcus mutans*, *Int. J. Antimicrob. Agents* 46 (2015) 174–182.
- [30] T. Singh, S. Das, V.G. Ramachandran, S. Wani, D. Shah, K.A. Maroof, A. Sharma, Distribution of integrons and phylogenetic groups among enteropathogenic *Escherichia coli* isolates from children <5 Years of age in Delhi, India, *Front. Microbiol.* 8 (2017) 561.
- [31] Clinical and Laboratory Standards Institute, Standards development policies and process, Available online: https://clsi.org/media/1874/s-001_standards_development_policies_and_processes.pdf. (Accessed 1 November 2019).
- [32] R.H. Lyles, C. Poindexter, A. Evans, M. Brown, C.R. Cooper, Nonlinear model-based estimates of IC50 for studies involving continuous therapeutic dose–response data, *Contemp. Clin. Trials* 29 (2008) 878–886.
- [33] X. Qin, D.M. Zerr, S.J. Weissman, J.A. Englund, D.M. Denno, E.J. Klein, P.I. Tarr, J. Kwong, J.R. Stapp, L.G. Tulloch, E. Galanakis, Prevalence and mechanisms of broad-spectrum β -lactam resistance in Enterobacteriaceae: a children's hospital experience, *Antimicrob. Agents Chemother.* 52 (2008) 3909–3914.
- [34] C. Levesque, L. Piche, C. Larose, P.H. Roy, PCR mapping of integrons reveals several novel combinations of resistance genes, *Antimicrob. Agents Chemother.* 39 (1995) 185–191.
- [35] S.F. Altschul, T.L. Madden, A.A. Schäffer, J. Zhang, Z. Zhang, W. Miller, D. J. Lipman, Gapped BLAST and PSI-BLAST: a new generation of protein database search programs, *Nucleic Acids Res.* 25 (1997) 3389–3402.
- [36] K. Arnold, L. Bordoli, J. Kopp, T. Schwede, The SWISS-MODEL workspace: a web-based environment for protein structure homology modelling, *Bioinformatics* 22 (2006) 195–201.
- [37] R.A. Laskowski, V.V. Chistyakov, J.M. Thornton, PDB sum more: new summaries and analyses of the known 3D structures of proteins and nucleic acids, *Nucleic Acids Res.* 33 (2005) D266–D268.
- [38] F.C. Bernstein, T.F. Koetzle, G.J. Williams, E.F. Meyer Jr., M.D. Brice, J.R. Rodgers, O. Kennard, T. Shimanouchi, M. Tasumi, The Protein Data Bank: a computer-based archival file for macromolecular structures, *Eur. J. Biochem.* 80 (1977) 319–324.
- [39] G.M. Sastry, M. Adzhigirey, T. Day, R. Annabhimoju, W. Sherman, Protein and ligand preparation: parameters, protocols, and influence on virtual screening enrichments, *J. Comput. Aided Mol. Des.* 27 (2013) 221–234.
- [40] N. Saitou, M. Nei, The neighbor-joining method: a new method for reconstructing phylogenetic trees, *Mol. Biol. Evol.* 4 (1987) 406–425.
- [41] K. Tamura, M. Nei, S. Kumar, Prospects for inferring very large phylogenies by using the neighbor-joining method, *Proc. Natl. Acad. Sci. Unit. States Am.* 101 (2004) 11030–11035.
- [42] K. Tamura, G. Stecher, D. Peterson, A. Filipski, S. Kumar, MEGA6: molecular evolutionary genetics analysis version 6.0, *Mol. Biol. Evol.* 30 (2013) 2725–2729.
- [43] WHO global strategy for containment of antimicrobial resistance, Available online: https://www.who.int/drugresistance/WHO_Global_Strategy.htm/en. (Accessed 10 October 2019).
- [44] C.J. von Wintersdorff, J. Penders, J.M. van Niekerk, N.D. Mills, S. Majumder, L. B. van Alphen, P.H. Savelkoul, P.F. Wolffs, Dissemination of antimicrobial resistance in microbial ecosystems through horizontal gene transfer, *Front. Microbiol.* 7 (2016) 173.
- [45] S.R. Partridge, S.M. Kwong, N. Firth, S.O. Jensen, Mobile genetic elements associated with antimicrobial resistance, *Clin. Microbiol. Rev.* 31 (2018).
- [46] A.M. Egorov, M.M. Ulyashova, M.Y. Rubtsova, Bacterial enzymes and antibiotic resistance, *Acta Naturae* 10 (2018) 33–48.
- [47] P.K. Mantravadi, K.A. Kalesh, R.C.J. Dobson, A.O. Hudson, A. Parthasarathy, The quest for novel antimicrobial compounds: emerging trends in Research, development, and technologies, *Antibiotics* (Berlin) 8 (2019).
- [48] P. Chaijenjawong, S.J. Foster, Desiccation tolerance in *Staphylococcus aureus*, *Arch. Microbiol.* 193 (2011) 125–135.
- [49] M. Tarchouna, A. Ferjani, S. Tilouche, M. Marzouk, J. Bouguila, L. Boughamouira, J. Boukadida, Screening at admission for carrier prevalence of multidrug-resistant organisms in a pediatric unit, *Am. J. Infect. Contr.* 42 (2014) 88–89.
- [50] M. Österblad, A. Hakanen, R. Manninen, T. Leisteuvio, R. Pelttonen, O. Meurman, P. Huovinen, P. Kotilainen, A between-species comparison of antimicrobial resistance in enterobacteria in fecal flora, *Antimicrob. Agents Chemother.* 44 (2000) 1479–1484.
- [51] P. Martínez-Freije, A.C. Fluit, F.J. Schmitz, J. Verhoef, M.E. Jones, Many class I integrons comprise distinct stable structures occurring in different species of Enterobacteriaceae isolated from widespread geographic regions in Europe, *Antimicrob. Agents Chemother.* 43 (1999) 686–689.
- [52] R. Banerjee, J. Strahilevitz, J.R. Johnson, P.P. Nagwekar, D.M. Schora, I. Shevrin, H. Du, L.R. Peterson, A. Robicsek, Predictors and molecular epidemiology of community-onset extended-spectrum β -lactamase-producing *Escherichia coli* infection in a Midwestern community, *Infect. Control Hosp. Epidemiol.* 34 (2013) 947–953.
- [53] T. Singh, P.K. Singh, S.A. Dar, S. Haque, N. Akhter, S. Das, Changing paradigm of antibiotic resistance amongst *Escherichia coli* isolates in Indian pediatric population, *PLoS One* 14 (2019), e0213850.
- [54] D.M. Zerr, X. Qin, A.P. Oron, A.L. Adler, D.J. Wolter, J.E. Berry, L. Hoffman, S. J. Weissman, Pediatric infection and intestinal carriage due to extended spectrum cephalosporin-resistant enterobacteriaceae, *Antimicrob. Agents Chemother.* 58 (2014) 3997–4004.
- [55] A.Q. Yaffee, Notes from the Field: verona integron-encoded metallo-beta-lactamase-producing carbapenem-resistant Enterobacteriaceae in a neonatal and adult intensive care unit—Kentucky, *MMWR Morb. Mortal. Wkly. Rep.* 65 (2015) 190, 2016.
- [56] K.K. Kumarasamy, M.A. Toleman, T.R. Walsh, J. Bagaria, F. Butt, R. Balakrishnan, U. Chaudhary, M. Doumith, C.G. Giske, S. Irfan, P. Krishnan, A.V. Kumar, S. Maharjan, S. Mushtaq, T. Noorie, D.L. Paterson, A. Pearson, C. Perry, R. Pike, B. Rao, U. Ray, J.B. Sarma, M. Sharma, E. Sheridan, M.A. Thirunarayan, J. Turton, S. Upadhyay, M. Warner, W. Welfare, D.M. Livermore, N. Woodford, Emergence of a new antibiotic resistance mechanism in India, Pakistan, and the UK: a molecular, biological, and epidemiological study, *Lancet Infect. Dis.* 10 (2010) 597–602.
- [57] M.L. Riccio, L. Pallecchi, R. Fontana, G.M. Rossolini, In70 of plasmid pAX 22, bla VIM-1-containing integron carrying a new aminoglycoside phosphotransferase gene cassette, *Antimicrob. Agents Chemother.* 45 (2001) 1249–1253.
- [58] P.J. Baudry, L. Mataseje, G.G. Zhanell, D.J. Hoban, M.R. Mulvey, Characterization of plasmids encoding CMY-2 AmpC β -lactamases from *Escherichia coli* in Canadian intensive care units, *Diagn. Microbiol. Infect. Dis.* 65 (2009) 379–383.
- [59] P. Nordmann, L. Poirel, M.A. Toleman, T.R. Walsh, Does broad-spectrum β -lactam resistance due to NDM-1 herald the end of the antibiotic era for treatment of infections caused by Gram-negative bacteria? *J. Antimicrob. Chemother.* 66 (2011) 689–692.
- [60] D. Seeliger, B.L. De Groot, Conformational transitions upon ligand binding: holotransformation prediction from apo conformations, *PLoS Comput. Biol.* 6 (2010), e1000634.
- [61] A. Dickson, Mapping the ligand binding landscape, *Biophys. J.* 115 (2018) 1707–1719.